

MATH-2111

Tutorial 1: Linear Systems, Vector Span & Parametric Forms

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Quick Review: Echelon Forms

1. Echelon vs. Reduced Echelon Form

- **Echelon Form (EF):** All zero rows are at the bottom; leading entries move to the right as you go down; all entries below a leading entry are zero.
- **Reduced Echelon Form (REF):** It is in EF, *plus* every leading entry is 1, and it is the *only* non-zero entry in its column.

Example of EF

$$\begin{bmatrix} \mathbf{2} & -3 & 2 & 1 \\ 0 & 0 & \mathbf{4} & 8 \\ 0 & 0 & 0 & 0 \end{bmatrix}$$

(Leading entries can be any non-zero number, only entries **below** are zero)

Example of REF

$$\begin{bmatrix} \mathbf{1} & -3 & 0 & 5 \\ 0 & 0 & \mathbf{1} & 2 \\ 0 & 0 & 0 & 0 \end{bmatrix}$$

(Leading entries must be **1**, and entries **above and below** are zero)

Quick Review: Core Theorems

2. The Fundamental Theorem of $Ax = b$

For an $m \times n$ matrix A , the following are logically equivalent:

- (a) For each $b \in \mathbb{R}^m$, the equation $Ax = b$ has a solution. $\Rightarrow Ax = x_1a_1 + x_2a_2 + \dots + x_na_n = b$
- (b) Each $b \in \mathbb{R}^m$ is a linear combination of the columns of A .
- (c) The columns of A span $\mathbb{R}^m$.
- (d) A has a pivot position in **every row**. (For $\mathbb{R}$)

$\Downarrow$
 $\text{Rank}(A) = m$

$$\left[\begin{array}{cc|c} 1 & 0 & b_1 \\ & 1 & b_2 \end{array} \right] \Rightarrow \begin{cases} x_1 = b_1 \\ x_2 = b_2 \end{cases}$$

$$\left[\begin{array}{cc|c} 1 & 0 & 2 \\ & 0 & 1 \end{array} \right] \Rightarrow \text{No soln!}$$

Quick Review: Existence vs. Uniqueness

Limit A as REF

Two Fundamental Questions for Linear Systems

Every linear system $Ax = b$ boils down to two independent questions:

① Existence (Is it consistent?)

- Algebraically: Is the last column of the augmented matrix a pivot column? If yes. No sin
- Geometrically: Is b in the Span of the columns of A ?



Span a linear space

② Uniqueness (Is there only one?)

- Algebraically: Are there any free variables? (Does every column of A have a pivot?)
- Geometrically: Are the columns of A linearly independent?



If a column has no pivot, corresponding variable is **free**!

$$[A|b] = \left[\begin{array}{ccc|c} 1 & 1 & 2 & 3 \\ 0 & 1 & 2 & 2 \\ 0 & 0 & 0 & 1 \end{array} \right] \Leftrightarrow \begin{cases} x_1 + x_2 + 2x_3 = 3 \\ x_2 + 2x_3 = 2 \\ x_3 = 1 \end{cases}$$

$$\left[\begin{array}{ccc|c} 1 & 1 & 2 & 3 \\ 0 & 1 & 2 & 2 \\ 0 & 0 & 0 & 1 \end{array} \right]$$

x_4 is free!

If yes, exists at least a column redundant

Quick Review: The Geometry of $Ax = b$

Deepening the Concept: A Vector Journey

Instead of just seeing $Ax = b$ as a set of static equations, view it as a **journey in vector space**:

$$x_1 \mathbf{a}_1 + x_2 \mathbf{a}_2 + \cdots + x_n \mathbf{a}_n = \mathbf{b}$$

Available directions:

destination:

$$\mathbf{a}_1 = \begin{bmatrix} 1 \\ 0 \\ 0 \end{bmatrix}, \mathbf{a}_2 = \begin{bmatrix} 0 \\ 1 \\ 0 \end{bmatrix}, \mathbf{b} = \begin{bmatrix} 2 \\ 3 \\ 5 \end{bmatrix}$$

- The columns of A ($\mathbf{a}_1, \dots, \mathbf{a}_n$) are your **available directions**.
- The variables $\mathbf{x} = (x_1, \dots, x_n)^T$ are the **steps (weights)** you take in each direction.
- The vector $\mathbf{b}$ is your **destination**.

Available directions:

$$\mathbf{a}_1 = \begin{bmatrix} 1 \\ 0 \end{bmatrix}, \mathbf{a}_2 = \begin{bmatrix} 0 \\ 1 \end{bmatrix}, \mathbf{a}_3 = \begin{bmatrix} 1 \\ 1 \end{bmatrix}, \mathbf{b} = \begin{bmatrix} 2 \\ 3 \end{bmatrix}$$

Conclusion: Solving $Ax = b$ simply means asking:

"How many steps must I take along each available direction to exactly reach my destination $\mathbf{b}$?"

True / False Concept Check

Determine if each statement is True or False:

- ① The homogeneous equation $A\mathbf{x} = \mathbf{0}$ has the trivial solution if and only if the equation has no free variables.
 $\mathbf{x} = \mathbf{0}$ $\nearrow$ always holds
- ② A 5×6 matrix can have at most 5 pivot columns.
- ③ If the equation $A\mathbf{x} = \mathbf{b}$ is consistent, then its solution set always represents a line or a plane passing through the origin. $\mathbf{b} = \mathbf{0}$
- ④ If A is an $m \times n$ matrix, and the columns of A do not span $\mathbb{R}^m$, then the equation $A\mathbf{x} = \mathbf{b}$ is inconsistent for all $\mathbf{b}$ in $\mathbb{R}^m$.

Answers & Justifications

- ① **False.** The homogeneous equation $A\mathbf{x} = \mathbf{0}$ *always* has the trivial solution ($\mathbf{x} = \mathbf{0}$), regardless of free variables. It has *only* the trivial solution if there are no free variables.
- ② **True.** A matrix can have at most one pivot per row and at most one pivot per column. Since there are only 5 rows, there can be at most 5 pivot positions in total.
- ③ **False.** The solution set of $A\mathbf{x} = \mathbf{b}$ is shifted away from the origin by a particular solution $\mathbf{p}$ (e.g., $\mathbf{x} = \mathbf{p} + t\mathbf{v}$). It only passes through the origin if $\mathbf{b} = \mathbf{0}$.
- ④ **False.** It just means there is *some* $\mathbf{b}$ for which the system is inconsistent. The system is still consistent for $\mathbf{b} = \mathbf{0}$ (trivial solution), and for any $\mathbf{b}$ that happens to lie within the actual span of the columns.

Problem 1: Solving Systems systematically

Problem 1

Use the row reduction algorithm to find the general solution of the following linear system:

$$x_1 + 2x_2 - 3x_3 = 4$$

$$2x_1 + x_2 + 3x_3 = 5$$

$$-x_1 + x_2 - 6x_3 = -1$$

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Step 1: Form the Augmented Matrix

$$\begin{bmatrix} 1 & 2 & -3 & 4 \\ 2 & 1 & 3 & 5 \\ -1 & 1 & -6 & -1 \end{bmatrix}$$

Solution to Problem 1

Step 2: Forward Phase (to Echelon Form)

$R_2 \rightarrow R_2 - 2R_1$ and $R_3 \rightarrow R_3 + R_1$:

$$\sim \begin{bmatrix} 1 & 2 & -3 & 4 \\ 0 & -3 & 9 & -3 \\ 0 & 3 & -9 & 3 \end{bmatrix}$$

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$$\sim \begin{bmatrix} 1 & 2 & -3 & 4 \\ 0 & -3 & 9 & -3 \\ 0 & 3 & -9 & 3 \end{bmatrix}$$

$R_2 \rightarrow -\frac{1}{3}R_2$, then $R_3 \rightarrow R_3 - 3R_2$:

$$\sim \begin{bmatrix} 1 & 2 & -3 & 4 \\ 0 & 1 & -3 & 1 \\ 0 & 0 & 0 & 0 \end{bmatrix}$$

Solution to Problem 1

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Step 3: Backward Phase (to Reduced Echelon Form)

$R_1 \rightarrow R_1 - 2R_2$:

$$\sim \begin{bmatrix} 1 & 0 & 3 & 2 \\ 0 & 1 & -3 & 1 \\ 0 & 0 & 0 & 0 \end{bmatrix}$$

Solution to Problem 1 (Cont.)

Step 4: Extract the General Solution

The system is consistent.

- **Pivot columns:** 1 and 2. $\implies$ Basic variables: x_1, x_2 .
- **Non-pivot column:** 3. $\implies$ Free variable: x_3 .

Re-writing the REF as equations:

$$x_1 + 3x_3 = 2 \implies x_1 = 2 - 3x_3$$

$$x_2 - 3x_3 = 1 \implies x_2 = 1 + 3x_3$$

Final Answer:

$$\begin{cases} x_1 = 2 - 3x_3 \\ x_2 = 1 + 3x_3 \\ x_3 \text{ is free} \end{cases}$$

Problem 2: Parameter Analysis

Problem 2

Given the augmented matrix of a linear system:

$$\left[\begin{array}{cc|c} 1 & h & 2 \\ 3 & 6 & k \end{array} \right]$$

Determine the values of h and k such that the system has: (a) A unique solution. (b) No solution. (c) Infinitely many solutions.

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Analysis: Apply $R_2 \rightarrow R_2 - 3R_1$:

$$\sim \begin{bmatrix} 1 & h & 2 \\ 0 & 6 - 3h & k - 6 \end{bmatrix}$$

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Analysis: Apply $R_2 \rightarrow R_2 - 3R_1$:

$$\sim \begin{bmatrix} 1 & h & 2 \\ 0 & 6 - 3h & k - 6 \end{bmatrix}$$

(c): $h=2, k=6$

- **(a) Unique solution:** Needs no free variables (pivot in every coefficient column).
 $\implies 6 - 3h \neq 0 \implies h \neq 2$ (and k can be any real number).
- **(b) No solution:** The rightmost column must be a pivot column $[0 \dots 0 \mid b \neq 0]$.
 $\implies 6 - 3h = 0$ and $k - 6 \neq 0 \implies h = 2$ and $k \neq 6$.

Problem 3: Geometry of the Span

Problem 3

Let $\mathbf{a}_1 = \begin{bmatrix} 1 \\ -2 \\ 0 \end{bmatrix}$, $\mathbf{a}_2 = \begin{bmatrix} 0 \\ 1 \\ 2 \end{bmatrix}$, $\mathbf{a}_3 = \begin{bmatrix} 3 \\ -4 \\ 4 \end{bmatrix}$, and $\mathbf{b} = \begin{bmatrix} 2 \\ -1 \\ 6 \end{bmatrix}$.

- (a) Determine if $\mathbf{b} \in \text{Span}\{\mathbf{a}_1, \mathbf{a}_2, \mathbf{a}_3\}$. If yes, write $\mathbf{b}$ as a linear combination.
- (b) What is the geometric object described by $\text{Span}\{\mathbf{a}_1, \mathbf{a}_2, \mathbf{a}_3\}$ in $\mathbb{R}^3$?

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Part (a): Solve $[\mathbf{a}_1 \ \mathbf{a}_2 \ \mathbf{a}_3 \mid \mathbf{b}]$:

$$\begin{bmatrix} 1 & 0 & 3 & 2 \\ -2 & 1 & -4 & -1 \\ 0 & 2 & 4 & 6 \end{bmatrix} \xrightarrow{R_2+2R_1} \begin{bmatrix} 1 & 0 & 3 & 2 \\ 0 & 1 & 2 & 3 \\ 0 & 2 & 4 & 6 \end{bmatrix} \xrightarrow{R_3-2R_2} \begin{bmatrix} 1 & 0 & 3 & 2 \\ 0 & 1 & 2 & 3 \\ 0 & 0 & 0 & 0 \end{bmatrix}$$

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Part (a): Solve $[\mathbf{a}_1 \ \mathbf{a}_2 \ \mathbf{a}_3 \mid \mathbf{b}]$:

$$\begin{bmatrix} 1 & 0 & 3 & 2 \\ -2 & 1 & -4 & -1 \\ 0 & 2 & 4 & 6 \end{bmatrix} \xrightarrow{R_2+2R_1} \begin{bmatrix} 1 & 0 & 3 & 2 \\ 0 & 1 & 2 & 3 \\ 0 & 2 & 4 & 6 \end{bmatrix} \xrightarrow{R_3-2R_2} \begin{bmatrix} 1 & 0 & 3 & 2 \\ 0 & 1 & 2 & 3 \\ 0 & 0 & 0 & 0 \end{bmatrix}$$

The system is consistent, so **$\mathbf{b}$ is in the span.**

Setting $x_3 = 0$, we get $x_1 = 2, x_2 = 3$. Hence, $\mathbf{b} = 2\mathbf{a}_1 + 3\mathbf{a}_2 + 0\mathbf{a}_3$.

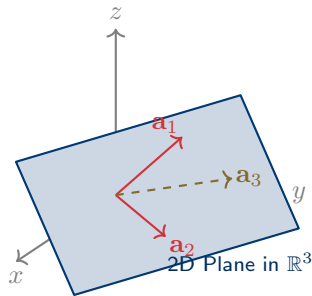
Problem 3 (Cont.): Geometric Interpretation

Part (b): Geometric object of $\text{Span}\{\mathbf{a}_1, \mathbf{a}_2, \mathbf{a}_3\}$

Look at the coefficient matrix after row reduction:

$$\begin{bmatrix} \mathbf{1} & 0 & 3 \\ 0 & \mathbf{1} & 2 \\ 0 & 0 & 0 \end{bmatrix}$$

- There are only **two pivot columns**.
- This means only two vectors (e.g., $\mathbf{a}_1, \mathbf{a}_2$) are linearly independent. $\mathbf{a}_3$ is redundant ($\mathbf{a}_3 = 3\mathbf{a}_1 + 2\mathbf{a}_2$).
- Since we have exactly two independent vectors in $\mathbb{R}^3$, their span generates a **2D plane passing through the origin** in 3-dimensional space.



Problem 4: Non-Homogeneous Systems

Problem 4

Find the general solution of $A\mathbf{x} = \mathbf{b}$ in **parametric vector form**, where:

$$A = \begin{bmatrix} 1 & -2 & 1 & 5 \\ -1 & 2 & 1 & 1 \\ 2 & -4 & 0 & 4 \end{bmatrix}, \quad \mathbf{b} = \begin{bmatrix} 5 \\ -1 \\ 6 \end{bmatrix}$$

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Row Reduce the Augmented Matrix:

$$\left[\begin{array}{cccc|c} 1 & -2 & 1 & 5 & 5 \\ -1 & 2 & 1 & 1 & -1 \\ 2 & -4 & 0 & 4 & 6 \end{array} \right] \xrightarrow[\text{then } R_3 - 2R_1]{R_2 + R_1} \left[\begin{array}{cccc|c} 1 & -2 & 1 & 5 & 5 \\ 0 & 0 & 2 & 6 & 4 \\ 0 & 0 & -2 & -6 & -4 \end{array} \right]$$

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Row Reduce the Augmented Matrix:

$$\begin{aligned} & \left[\begin{array}{ccccc} 1 & -2 & 1 & 5 & 5 \\ -1 & 2 & 1 & 1 & -1 \\ 2 & -4 & 0 & 4 & 6 \end{array} \right] \xrightarrow[\text{then } R_3 - 2R_1]{R_2 + R_1} \left[\begin{array}{ccccc} 1 & -2 & 1 & 5 & 5 \\ 0 & 0 & 2 & 6 & 4 \\ 0 & 0 & -2 & -6 & -4 \end{array} \right] \\ & \xrightarrow[\begin{smallmatrix} R_3 + R_2 \\ \frac{1}{2}R_2 \end{smallmatrix}]{R_1 - R_2} \left[\begin{array}{ccccc} 1 & -2 & 1 & 5 & 5 \\ 0 & 0 & 1 & 3 & 2 \\ 0 & 0 & 0 & 0 & 0 \end{array} \right] \xrightarrow{R_1 - R_2} \left[\begin{array}{ccccc} \color{red}{1} & -2 & 0 & 2 & 3 \\ 0 & 0 & \color{red}{1} & 3 & 2 \\ 0 & 0 & 0 & 0 & 0 \end{array} \right] \text{ (REF)} \end{aligned}$$

Solution to Problem 4 (Cont.)

Translate REF to Parametric Vector Form:

$$x_1 - 2x_2 + 2x_4 = 3 \implies x_1 = 3 + 2x_2 - 2x_4$$

$$x_3 + 3x_4 = 2 \implies x_3 = 2 - 3x_4$$

Solution to Problem 4 (Cont.)

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$$x_3 + 3x_4 = 2 \implies x_3 = 2 - 3x_4$$

Assemble the vector $\mathbf{x}$:

$$\mathbf{x} = \begin{bmatrix} x_1 \\ x_2 \\ x_3 \\ x_4 \end{bmatrix} = \begin{bmatrix} 3 + 2x_2 - 2x_4 \\ x_2 \\ 2 - 3x_4 \\ x_4 \end{bmatrix} = \begin{bmatrix} 3 \\ 0 \\ 2 \\ 0 \end{bmatrix} + x_2 \begin{bmatrix} 2 \\ 1 \\ 0 \\ 0 \end{bmatrix} + x_4 \begin{bmatrix} -2 \\ 0 \\ -3 \\ 1 \end{bmatrix}$$

Solution to Problem 4 (Cont.)

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Theorem Connection

$$\mathbf{x} = \mathbf{p} + x_2 \mathbf{v}_1 + x_4 \mathbf{v}_2$$

The solution set is a plane in $\mathbb{R}^4$. The vector $\mathbf{p} = (3, 0, 2, 0)^T$ is a *particular solution* to $A\mathbf{x} = \mathbf{b}$, while $x_2 \mathbf{v}_1 + x_4 \mathbf{v}_2$ represents the general solution to the homogeneous equation $A\mathbf{x} = \mathbf{0}$.

Problem 5: Synthesizing Concepts

Problem 5: Span and Consistency

Let $\mathbf{v}_1 = \begin{bmatrix} 1 \\ 2 \\ -1 \end{bmatrix}$, $\mathbf{v}_2 = \begin{bmatrix} -2 \\ -3 \\ 4 \end{bmatrix}$, $\mathbf{v}_3 = \begin{bmatrix} 0 \\ 1 \\ 2 \end{bmatrix}$, and $A = [\mathbf{v}_1 \ \mathbf{v}_2 \ \mathbf{v}_3]$.

- (a) Find an algebraic condition on b_1, b_2, b_3 such that an arbitrary vector $\mathbf{b} = \begin{bmatrix} b_1 \\ b_2 \\ b_3 \end{bmatrix}$ is in $\text{Span}\{\mathbf{v}_1, \mathbf{v}_2, \mathbf{v}_3\}$.
- (b) If $\mathbf{b} = \begin{bmatrix} 1 \\ 3 \\ 1 \end{bmatrix}$, verify it is in the span, and express the general solution to $A\mathbf{x} = \mathbf{b}$ in parametric vector form.

Solution to Problem 5 (Part a)

(a) Set up the augmented matrix $[A \mid \mathbf{b}]$:

$$\begin{bmatrix} 1 & -2 & 0 & b_1 \\ 2 & -3 & 1 & b_2 \\ -1 & 4 & 2 & b_3 \end{bmatrix}$$

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(a) Set up the augmented matrix $[A \mid \mathbf{b}]$:

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Apply $R_2 \rightarrow R_2 - 2R_1$ and $R_3 \rightarrow R_3 + R_1$:

$$\sim \begin{bmatrix} 1 & -2 & 0 & b_1 \\ 0 & 1 & 1 & b_2 - 2b_1 \\ 0 & 2 & 2 & b_3 + b_1 \end{bmatrix}$$

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Apply $R_3 \rightarrow R_3 - 2R_2$:

$$\sim \begin{bmatrix} 1 & -2 & 0 & b_1 \\ 0 & 1 & 1 & b_2 - 2b_1 \\ 0 & 0 & 0 & (b_3 + b_1) - 2(b_2 - 2b_1) \end{bmatrix}$$

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For the system to be consistent, the last entry must be zero.

Condition: $\boxed{5b_1 - 2b_2 + b_3 = 0}$. (This describes a plane in $\mathbb{R}^3$).

Solution to Problem 5 (Part b)

(b) Evaluate with specific $\mathbf{b}$:

Given $\mathbf{b} = \begin{bmatrix} 1 \\ 3 \\ 1 \end{bmatrix}$. Check the condition: $5(1) - 2(3) + 1 = 0$.

Since the condition holds, **it is in the span.**

Solution to Problem 5 (Part b)

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Given $\mathbf{b} = \begin{bmatrix} 1 \\ 3 \\ 1 \end{bmatrix}$. Check the condition: $5(1) - 2(3) + 1 = 0$.

Since the condition holds, **it is in the span.**

Substitute $b_1 = 1, b_2 = 3, b_3 = 1$ into the echelon form from part (a):

$$\begin{bmatrix} 1 & -2 & 0 & 1 \\ 0 & 1 & 1 & 3 - 2(1) \\ 0 & 0 & 0 & 0 \end{bmatrix} = \begin{bmatrix} 1 & -2 & 0 & 1 \\ 0 & 1 & 1 & 1 \\ 0 & 0 & 0 & 0 \end{bmatrix}$$

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Row reduce to REF ($R_1 \rightarrow R_1 + 2R_2$):

$$\sim \begin{bmatrix} 1 & 0 & 2 & 3 \\ 0 & 1 & 1 & 1 \\ 0 & 0 & 0 & 0 \end{bmatrix} \implies \begin{cases} x_1 = 3 - 2x_3 \\ x_2 = 1 - x_3 \\ x_3 \text{ is free} \end{cases}$$

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Parametric vector form: